

Highlights

- Predictive HVAC states drive adaptive event-triggered policy execution.
- Multi-scale scores track load changes across building operating timescales.
- Local-global thresholds adapt without labeled event data or fixed rules.
- ET-PRL cuts policy executions by 53.54% and chiller energy use by 2.13%.
- Triggers are 2.02 times denser during high load variation than stable periods.